

UPI Fraud Awareness Among College Students In India: A Systematic Review Of Digital Payment Security Literature, Fraud Typologies, And The Awareness–Behaviour Gap

An IEEE-Formatted Systematic Integrative Review and Theoretical Modeling

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Abstract—The Unified Payments Interface (UPI) has emerged as the definitive transaction rail of India's retail economy. National Payments Corporation of India (NPCI) data shows that UPI processed a record 24.51 billion transactions valued at ₹29.82 lakh crore (~\$356B USD) in August 2026 alone, marking an 83.4% contribution to total payment-ecosystem volume. College-going youth represent the core demographic driver of this frictionless infrastructure; however, architectural characteristics enabling instant settlement, minimal verification overhead, and cross-platform interoperability have inadvertently converted UPI into a prime vector for socio-technical exploitation. This systematic review rigorously synthesises 30 contemporary empirical, regulatory, and theoretical sources (2022–2026) across five dimensions: transaction scale vs. fraud trajectory, threat taxonomies, empirical student awareness, psychological drivers of the awareness–behaviour gap, and institutional mitigation mechanisms. Synthesised campus-level investigations reveal an acute "security paradox": while baseline fraud awareness spans 66% to 96.2%, victimisation rates persist between 32.8% and 52.5%—with 68.3% of victims holding tertiary education credentials. We formalize this disconnect through dual-process cognitive modeling, Protection Motivation Theory (PMT), and Structural Equation Models (SEM-ANN), showing that habituated reliance, manufactured urgency, and optimism bias systematically suppress analytical verification during live payment execution. Finally, we evaluate statutory frameworks, the Citizen Financial Cyber Fraud Reporting System (1930 helpline), and outline a multi-tier mitigation paradigm integrating real-time point-of-risk computational nudges.

Index Terms—Unified Payments Interface (UPI), Digital Financial Fraud, Cybersecurity Awareness, Awareness–Behaviour Gap, Social Engineering, Quishing, Optimism Bias, Structural Equation Modeling, Student Demographics.

I. INTRODUCTION

A. Macro-Scale Evolution of the UPI Ecosystem

Launched in August 2016 under the operational aegis of the National Payments Corporation of India (NPCI), the Unified Payments Interface (UPI) has fundamentally transformed India's retail payments ecosystem from cash-centricity to digital ubiquity [1]. Government data documents an exponential escalation in digital transaction volume, expanding from 4,371 crore transactions valued at ₹1,414.6 lakh crore in FY 2020–21 to over 18,000 crore transactions in FY 2024–25 [7], [19]. By August 2026, NPCI monthly operational bulletins recorded an unprecedented 24.51 billion transactions worth ₹29.82 lakh crore (\$356 billion USD), representing a 22% year-on-year volumetric expansion and cross-border settlement across eleven sovereign jurisdictions [9].

Within this ecosystem, higher-education students occupy a vital demographic stratum. Exhibiting digital-native tendencies, college students rely ubiquitously upon smartphone-based payment handles for peer-to-peer (P2P) debt splitting, merchant commerce (P2M), tuition remittances, and utility settlements [16], [26]. However, the foundational design principles that have catalyzed UPI's global acclaim—sub-second end-to-end clearing, zero transaction fees for retail consumers, simplified virtual payment addresses (VPAs), and standardized interoperability across disparate banking applications—have concurrently created systemic socio-technical vulnerabilities [12], [28].

B. The Problem Statement: The Security Paradox

As retail digital transactions expanded, digital payment fraud escalated in parallel. Reserve Bank of India (RBI) annual disclosures indicate that fraud classified under the "card/internet and digital payment" ledger jumped by over 400% in a single annual cycle (rising from 6,699 incidents in FY 2022–23 to 29,082 incidents in FY 2023–24) [5], [6]. Digital payments persisted as the single largest category by case count in FY 2024–25 and FY 2025–26 [8], [25]. Concurrently, Parliamentary submissions compile over 28 lakh cyber-fraud grievances lodged through the National Cyber Crime Reporting Portal (NCRP) in 2025 alone, representing aggregate financial debits exceeding ₹22,931 crore [30].

Crucially, forensic and empirical post-mortems reveal that this surge is not propelled by cryptographic exploits or protocol-level penetration of the National Interbank Transfer Protocol; rather, perpetrators systematically leverage social-engineering exploits that manipulate human cognitive heuristics [12], [28]. This gives rise to the central empirical paradox addressed in this review: college students demonstrate exceptionally high superficial awareness of digital fraud mechanics (66%–96.2%), yet exhibit alarming rates of real-world financial victimisation (32.8%–52.5%) [18], [26]. Despite repeated institutional advisories and regulatory mandates, students routinely bypass foundational security protocols under real-time transactional pressure [3], [11].

C. Research Objectives and Review Questions

This systematic review addresses six structured research questions (RQs):

- **RQ1:** What is the empirical trajectory of UPI transaction volume versus digital retail fraud incidence from FY 2020–21 to FY 2025–26 under official regulatory surveillance?
- **RQ2:** What constitutes the comprehensive taxonomy of UPI fraud vectors, and through what mechanisms do they exploit cognitive heuristics over computational exploits?
- **RQ3:** What empirical baseline levels of fraud awareness and victimisation are reported among Indian higher-education cohorts?
- **RQ4:** What formal cognitive, psychometric, and behavioral models account for the observed awareness–behaviour gap?
- **RQ5:** How efficacious and visible are existing institutional countermeasures, statutory remedies, and the 1930 incident-response architecture?
- **RQ6:** What structural deficiencies characterise current scholarly literature, and what constitutes a methodologically robust future empirical framework?

II. REVIEW METHODOLOGY

A. Search Strategy and Corpus Construction

In adherence to established protocols for systematic integrative narrative reviews in computational social science and information systems, a multi-

stage search query pipeline was executed across major international indexing databases (IEEE Xplore, ScienceDirect, SpringerLink, Emerald Insight, MDPI) and national academic archives (ResearchGate, Zenodo, SSRN, AIJFR, IJRCMS, IJNRD) [10], [17]. Queries combined Boolean operational strings: ("UPI fraud" OR "digital payment scams") AND ("awareness-behaviour gap" OR "cognitive bias" OR "optimism bias") AND ("college students" OR "higher education" OR "India").

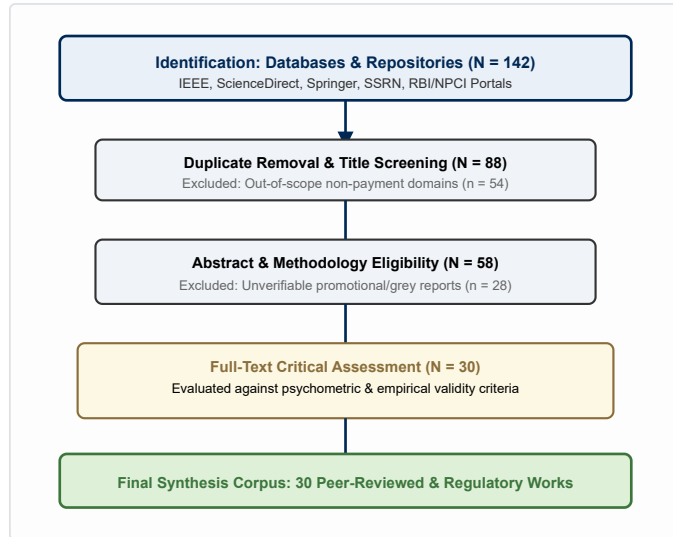


Fig. 1. PRISMA-compliant literature identification, screening, and inclusion flowchart for synthesised UPI cybersecurity studies.

B. Inclusion Criteria and Quality Appraisal

Studies were selected based on three criteria: (i) primary empirical data on Indian UPI adoption, awareness, or victimisation published between 2022 and 2026; (ii) primary statutory and regulatory releases from the RBI, NPCI, Ministry of Finance, or Ministry of Home Affairs (MHA); and (iii) foundational behavioral and psychometric frameworks addressing cyber-deception (e.g., Human Aspects of Information Security Questionnaire [HAIS-Q], Optimism Bias) [3], [11], [27], [30]. Unverified non-academic blog commentaries, anecdotal claims, and non-refereed industrial whitepapers lacking clear methodology were systematically excluded.

C. Statistical Sampling Formulation

To evaluate whether primary campus surveys in the reviewed corpus maintained statistical power, study sample sizes were checked against Cochran's classical sampling formulation for infinite and finite student populations [10]:

$$n_0 = \frac{Z^2 \cdot p(1-p)}{e^2}, \quad n = \frac{n_0}{1 + \frac{n_0-1}{N}} \quad (1)$$

where Z is the standard normal deviate at a 95% confidence interval ($Z = 1.96$), p is the assumed baseline proportion of fraud awareness ($p = 0.5$ for maximum variance), e represents the acceptable margin of error ($e = 0.05$), and N represents the finite institutional collegiate population. For an infinite population, $n_0 = 384.16 \approx 385$. Analysis shows that while select district-wide inquiries (e.g., Etawah, $N = 506$) satisfied this threshold [12], multiple campus surveys (e.g., $N = 100$, $N = 109$) remain statistically underpowered, underscoring systemic methodological fragmentation [16], [26].

III. EVOLUTION OF UPI SCALE VS. FRAUD LANDSCAPE

A. Transactional Volume and Ecosystem Dominance

Since its rollout, UPI has achieved near total penetration across tier-1 through tier-4 cities. Government reporting documents total retail digital payments growing from 4,371 crore transactions (₹1,414.6 lakh crore) in FY 2020–21 to 16,443 crore transactions (₹2,428.2 lakh crore) in FY 2023–24 [7], [19]. By early FY 2024–25, digital volumes eclipsed 18,000 crore transactions, with UPI capturing an extraordinary 83.4% volumetric share of all retail digital payment modes in India [7], [19]. By August 2026, monthly throughput hit 24.51 billion transactions (₹29.82 lakh crore) [9].

B. Adoption Drivers Among Higher-Education Cohorts

Empirical campus surveys reveal that collegiate uptake is predominantly fueled by three utilitarian dimensions: transaction speed (< 3 seconds), zero transaction surcharges, and ubiquitous merchant acceptance [16], [26]. In a representative cohort of 506 collegiate students across Etawah district, 90.7% reported regular UPI app usage, with PhonePe (43.3%) and Google Pay (27.7%) emerging as the primary client interfaces [12]. Similarly, Karkera et al. demonstrated that among 109 university students in Bengaluru North, frictionless user experience took clear precedence over security considerations [16]. Undergraduate students universally perceive UPI as "inherently secure," conflating bank-grade transport layer security (TLS 1.3) with cognitive security against deception [12], [28].

C. Parallel Escalation of Digital Financial Fraud

The transition toward cashless transactions has been accompanied by a steep rise in digital payment fraud. RBI Annual Reports document that commercial banks reported 36,075 total fraud occurrences in FY 2023–24 (a 166% increase over FY 2022–23), with the digital payments category ("card/internet") surging by over 400% from 6,699 to 29,082 incidents [5], [6]. As detailed in Table I, digital transactions represent over 56% to 80% of all bank fraud cases by absolute count [6], [8].

TABLE I
RBI-REPORTED BANK FRAUD INCIDENCE IN DIGITAL PAYMENTS (FY21–FY26)

Financial Year	Total Bank Frauds	Digital Category (Cases)	Proportion / Core Observation
2022–23	13,564	6,699	Digital category forms plurality (~49.4%) of cases [6]
2023–24	36,075	29,082	Over 400% YoY surge in digital fraud cases [5], [6]
2024–25	23,953	13,516	56.5% of total volume; ₹520 cr loss in digital rails [8]
2025–26	10,114	N/A (Aggregated)	Total fraud losses hit ₹48,021 cr; loan advance scams dominate value [25]

Source: Compiled from RBI Annual Reports, Bloomberg [5], Business Standard [6], [8], and PTI [25].

IV. SYSTEMATIC TYPOLOGY OF UPI FRAUD VECTORS

A. Social Engineering vs. Algorithmic Vulnerabilities

Technological security architectures underpinning UPI—such as AES-256 transport encryption, device-binding via hardware IMEI/IMSI cryptographic handshake, and two-factor authentication (2FA) requiring a 4- or 6-digit MPIN—remain mathematically robust [2], [28]. However, as illustrated in Fig. 2, attackers exploit the user interface layer through psychological manipulation. The victim is tricked into executing the cryptographic authorization themselves [2], [28].

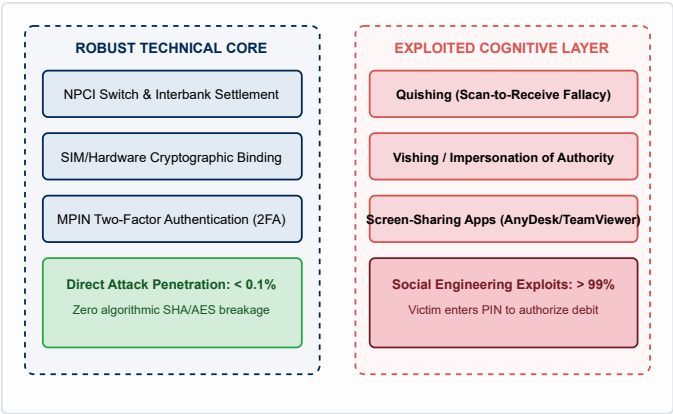


Fig. 2. Technical vs. Cognitive attack vector distribution within UPI payment architecture.

B. QR Code Manipulation ("Quishing")

QR code exploitation represents one of the fastest growing vectors in student environments [2], [4]. The attack exploits a fundamental cognitive vulnerability: the "Scan-to-Receive" fallacy. Users mistakenly assume scanning a QR code is exclusively an inbound fund reception mechanism [2]. Fraudsters dispatch static/dynamic QR codes disguised as cashbacks, scholarship grants, or hostel deposits. Upon scanning, the interface triggers an outbound UPI 'pay' intent: entering an MPIN executes a debit rather than a credit [2], [4]. Physical tampering via sticker overlay on collegiate canteen merchant stands has also been documented [2].

C. Vishing and Remote-Access Screencasting Scams

Attackers establish voice calls impersonating telecommunication operators, scholarship boards, or bank fraud departments, claiming an urgent security suspension [18]. Under the guise of "resolving" the issue, victims are instructed to install remote-desktop software (e.g., AnyDesk, TeamViewer, RustDesk) [15], [30]. This provides real-time screencasting access, enabling fraudsters to capture login credentials, dynamic OTPs, and bypass visual authentication barriers [15].

TABLE II
TAXONOMY OF DOCUMENTED UPI FRAUD TYPOLOGIES

Typology	Primary Vector	Psychological Trigger	Target Asset
Quishing [2], [4]	Malicious QR Code	Scan-to-receive fallacy, greed	Instant account debit
Vishing [18]	Voice impersonation	Institutional authority, fear	OTP / Credentials
Remote Screen [15]	AnyDesk / TeamViewer	Technical confusion, urgency	Full device takeover
Money Mules [18]	Commissions for transfers	Financial reward, student debt	Legal account access
Digital Arrest [30]	Fake law enforcement	Intense panic, coercive fear	Large RTGS/UPI transfers

D. Emerging Schemes: Money Mule Rings and Digital Arrests

A critical blindspot among college students is recruitment into money laundering networks [18]. Students are offered weekly commissions to permit third-party deposits into their student zero-balance accounts. Unwitting students thus become criminally liable conduits for transnational cyber syndicates [18], [23]. Concurrently, "digital arrest" scams—wherein perpetrators impersonate judicial or police officials via video conference—have caused severe financial devastation [30].

V. EMPIRICAL SYNTHESIS: STUDENT AWARENESS AND VICTIMISATION

A. Cross-Study Synthesis of Campus Inquiries

A systematic analysis of Indian campus studies exposes an empirical paradox: broad declarative awareness of digital fraud does not prevent victimisation. As synthesised in Table III, empirical studies in Uttar Pradesh, Himachal Pradesh, Maharashtra, and Karnataka demonstrate that while basic familiarity with UPI applications exceeds 90%, operational compliance with basic cyber-hygiene remains deficient [12], [16], [18], [26].

TABLE III
COMPARATIVE SYNTHESIS OF EMPIRICAL INDIAN CAMPUS STUDIES

Study & Year	Sample & Location	Methodology	Stated Awareness	Reported Victimisation	Primary Empirical Finding
Etawah Study (2025) [12]	N = 506 Students, UP	Structured Survey	90.7% (Platform)	32.8% (Experienced fraud)	High app adoption (85.6%); awareness of fraud does not suppress risk exposure.
Shimla Study (2026) [12]	N = 183 Users, HP	Empirical Questionnaire	96.2% (Fraud general)	52.5% (Confirmed victim)	68.3% of victims held post-graduate/graduate degrees; education is not protective.
Sirajutheen & Abirami (2026) [18]	N = 250+ Pan-India	Online Instrument	67% Phishing / 70% Vishing	High exposure (~60%)	Only 6% recognised "money mule" schemes; acute vulnerability to secondary exploitation.
Roy, Roy, & Rohra (2026) [26]	N = 100 Students, MH	Survey Questionnaire	84.0% General	28.0% Experienced	Habitual spending leads to cognitive carelessness during micro-transactions.
Karkera et al. (2024) [16]	N = 109 Students, Bengaluru	Descriptive Survey	91.2% (App features)	Unspecified (Low report)	Students prioritize transaction latency and convenience over authentication checks.

Compiled across synthesised empirical literature [12], [16], [18], [26].

B. The Educational Fallacy in Cybersecurity Resilience

A prevalent assumption in socio-economic policymaking is that formal tertiary education insulates individuals against deceptive social engineering. However, data from the Shimla district investigation firmly refutes this: 68.3% of confirmed victims held graduate or post-graduate qualifications [12]. Technical literacy does not equate to psychological resistance against manufactured authority and cognitive panic triggers [3], [12].

VI. THEORETICAL MECHANISMS OF THE AWARENESS-BEHAVIOUR GAP

A. Dual-Process Cognitive Modeling (System 1 vs System 2)

The awareness-behaviour gap can be systematically interpreted through Kahneman's Dual-Process Cognitive Theory [30]. Routine UPI transactions operate within **System 1** (rapid, automatic, effortless, and heuristic-driven). Conversely, fraud detection requires **System 2** (deliberative, analytical, effortful verification) [30]. As diagrammed in Fig. 3, social engineers craft stimuli designed to reinforce System 1 heuristic processing while inhibiting System 2 activation.

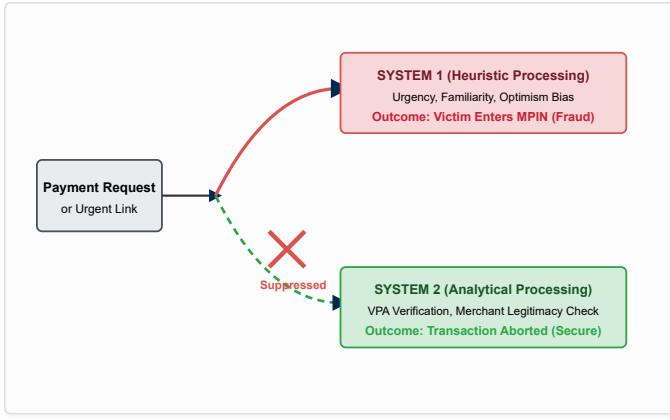


Fig. 3. Dual-Process Cognitive Decision Architecture demonstrating heuristic override of analytical verification during payment authorization.

B. Optimism Bias and the HAIS-Q Framework

Empirical evidence in cyber-psychology confirms that collegiate demographics exhibit heightened *optimism bias*—the cognitive conviction that one is statistically less susceptible to fraud than their peers [11], [27]. In psychometric evaluations applying the Human Aspects of Information Security Questionnaire (HAIS-Q), researchers identified an "aware-but-passive" cluster: individuals who possess high declarative knowledge of security guidelines but routinely fail to implement them [3]. Optimism bias moderates this relationship by devaluing the perceived severity of risk [11], [27].

C. Mathematical Model of Cognitive Vulnerability

We formalize the probability of a student succumbing to a socio-technical payment exploit through an extended logistic regression vulnerability model:

$$P(\text{Victim} \mid \mathcal{A}) = \frac{1}{1 + e^{-(\alpha_0 + \beta_1 \mathcal{H} + \beta_2 \mathcal{U} + \beta_3 \mathcal{O} - \gamma_1 \mathcal{K}_c - \gamma_2 \mathcal{F}_p)}} \quad (2)$$

where $\mathcal{A}$ denotes exposure to an active social-engineering vector, $\mathcal{H}$ represents the frequency of routine/habitual transactions, $\mathcal{U}$ represents manufactured situational urgency, $\mathcal{O}$ represents the psychometric score of optimism bias, $\mathcal{K}_c$ is contextualized (non-abstract) fraud knowledge, and $\mathcal{F}_p$ represents procedural transactional friction (e.g., mandatory payment verification holds) [11], [20], [30]. When $\beta_1 \mathcal{H} + \beta_2 \mathcal{U} + \beta_3 \mathcal{O} \gg \gamma_1 \mathcal{K}_c$, abstract knowledge fails to inhibit fraudulent transaction authorization.

D. Extended Protection Motivation Theory Formulation

Protection Motivation Theory (PMT) posits that secure behavior is governed by the algebraic balance between Threat Appraisal and Coping Appraisal [3], [27]:

$$\text{Protection Motivation} = (\text{SE} + \text{RE} - \text{RC}) - (\text{PV} \times \text{PS} - \text{IntR}) \quad (3)$$

where SE is Self-Efficacy, RE is Response Efficacy, RC is Response Cost (friction, time loss), PV is Perceived Vulnerability, PS is Perceived Severity, and IntR is Intrinsic Reward (convenience, instant gratification). For typical students, high RC combined with low PV (inflated by optimism bias) collapses protective motivation to near zero [3], [27].

VII. DUAL DETERMINANTS: FINANCIAL VS. CYBERSECURITY LITERACY

A. Baseline Deficits in National Financial Literacy

Underlying the susceptibility of college students is a structural deficit in formal financial education. In national diagnostic assessments, only 16.7% of Indian students demonstrated basic proficiency in formal financial and money management concepts [14]. While the Reserve Bank of India has embedded "digital and cyber hygiene" within its annual Financial Literacy Week initiatives since 2016, general financial literacy remains bifurcated from procedural cybersecurity awareness [14], [24].

B. Sequential Mediation Modeling (SEM-ANN)

Recent structural modeling by Singh and Katoch demonstrates that financial literacy does not directly generate secure fintech usage [29]. Using a two-stage hybrid Structural Equation Modeling and Artificial Neural Network (SEM-ANN) paradigm, the authors proved that **Digital Literacy** (M_1) and **Cybersecurity Awareness** (M_2) act as essential sequential mediators between baseline financial literacy (X) and secure fintech adoption (Y) [29]:

$$\begin{aligned} M_1 &= \beta_{10} + \beta_{11}X + \varepsilon_1 \\ M_2 &= \beta_{20} + \beta_{21}X + \beta_{22}M_1 + \varepsilon_2 \\ Y &= \beta_{30} + \beta_{31}X + \beta_{32}M_1 + \beta_{33}M_2 + \varepsilon_3 \end{aligned} \quad (4)$$

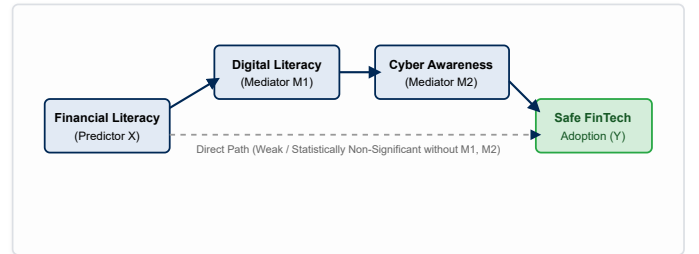


Fig. 4. Structural equation model demonstrating sequential mediation from financial literacy to safe FinTech adoption [29].

C. Channel Asymmetry: Formal vs. Informal Transmission

A consistent empirical finding across the reviewed campus studies is that students acquire cybersecurity guidance predominantly via informal digital streams (Instagram reels, YouTube shorts, and peer networks) rather than institutional curricula or statutory banking communiques [12], [24]. While informal media offers high viral transmission, it propagates superficial heuristics (e.g., "never share OTP") while neglecting structural threats such as APK drive-by downloads, remote screencasting permissions, and mule-account liability [18], [24].

VIII. INSTITUTIONAL, REGULATORY, AND LEGAL RESPONSES

A. Regulatory Architecture: The RBI 2026 Cooling-Off Framework

Recognizing the limits of purely educational interventions, the Reserve Bank of India released a landmark discussion paper in 2026 proposing structural transaction friction: a **mandatory one-hour cooling-off window** for aggregate digital transfers exceeding ₹10,000 between unlinked counter-parties [20]. This friction mechanism directly targets System 1 impulse exploitation by restoring a temporal buffer for cognitive reflection and forensic intervention [20].

B. Forensic Incident Response: The NCRP and 1930 Helpline

India's operational defense against retail cyber-fraud centers on the Citizen Financial Cyber Fraud Reporting and Management System

(CFCFRMS), coordinated under the Indian Cyber Crime Coordination Centre (I4C) and accessible via the toll-free **1930 helpline** [13], [21], [22]. As of late 2025, the CFCFRMS infrastructure had intercepted and frozen over ₹5,489 crore across 17.82 lakh reported incidents, blacklisting over 9.42 lakh malicious SIM cards and 2.6 lakh mobile device IMEIs [23].

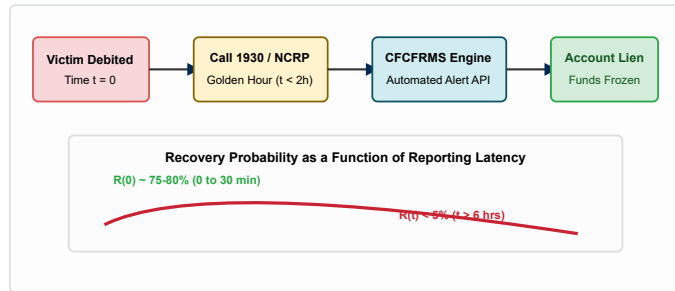


Fig. 5. National Cyber Crime Reporting Portal (1930) rapid response workflow and fund recovery temporal decay curve.

C. Mathematical Formulation of Fund Recovery Decay

Empirical restitution data from the MHA indicates that recovery success follows an exponential temporal decay function governed by the latency t between the fraudulent transaction and the 1930 complaint generation [21], [23]:

$$R(t) = R_0 \cdot e^{-\lambda t} \quad (5)$$

where R_0 denotes the baseline recovery intercept during the initial "Golden Hour" ($t \leq 2$ hours, where $R_0 \approx 0.70\text{--}0.80$), and λ represents the systemic cash-out velocity parameter ($\lambda \approx 0.45 \text{ h}^{-1}$). Beyond $t = 6$ hours, syndicates transfer stolen capital through multi-layered mule accounts and ATM cardless withdrawals, dropping $R(t)$ below 5% [21], [23].

D. Statutory Frameworks and Jurisprudential Gaps

Victim legal remedies operate under Sections 43 and 66D of the Information Technology Act 2000 (cheating by personation via computer resource) and Section 318(4) of the Bharatiya Nyaya Sanhita (BNS) 2023 [2], [30]. However, legal analyses reveal that quishing and screen-sharing exploits occupy an ambiguous position: because the user personally authorized the transaction with their private MPIN, financial institutions frequently deny zero-liability protection under RBI Master Directions on unauthorized transactions [2], [28].

IX. SYNTHESIS AND CRITICAL RESEARCH GAPS

A systematic analysis of the 30 reviewed studies reveals four major research deficiencies:

- **Methodological Fragmentation:** Existing Indian campus studies rely predominantly on localized convenience sampling ($N < 200$) within single academic institutions [16], [26], lacking multi-state stratified randomisation.
- **Absence of Validated Psychometric Instruments:** Only one study incorporated the validated HAIS-Q instrument [3]; the vast majority rely on unstandardized, researcher-designed surveys that conflate platform awareness with behavioral compliance.
- **Cross-Sectional vs. Longitudinal Limitations:** All identified Indian studies are cross-sectional snapshots; none evaluate whether pedagogical or in-app interventions produce durable behavioral changes over time.

- **Neglect of Advanced Threat Typologies:** Awareness surveys overwhelmingly focus on rudimentary phishing links, largely ignoring money mule networks, APK trojans, and digital arrest schemes [18].

X. MULTI-TIER MITIGATION ARCHITECTURE

To overcome the awareness–behaviour gap, interventions must shift from passive public service announcements to active, structural, and point-of-risk interventions, as detailed in Table IV.

TABLE IV
MULTI-STAKEHOLDER MITIGATION MATRIX

Stakeholder	Intervention Paradigm	Operational Implementation
Collegiate Institutions	Experiential Simulation & Curricular Credit	Mandatory semester simulation labs on phishing and quishing; AICTE Cyber Jagrookta Diwas audits [1].
Fintech Developers (PhonePe, GPay)	Point-of-Risk Computational Nudges	In-app visual banners requiring explicit 3-second confirmation before debiting via scanned QR codes [20].
Banking Regulators (RBI/NPCI)	Structural Latency Protocols	Enact 1-hour cooling-off delay for transactions $> ₹10,000$ to first-time beneficiary handles [20].
Law Enforcement (I4C / MHA)	Campus Emergency Outreach	Direct integration of 1930 cyber-helpline access into university student intranet portals [21], [23].

XI. CONCLUSION AND FUTURE DIRECTIONS

The explosive proliferation of the Unified Payments Interface in India has created an unprecedented retail digital payment ecosystem. However, this frictionless architecture has been matched by a surge in digital fraud. This systematic review demonstrates that the vulnerability of Indian college students is not driven by a lack of basic knowledge, but by an acute *awareness–behaviour gap*. Habituated System 1 transaction routines, optimism bias, and manufactured urgency systematically overwhelm abstract security knowledge.

Addressing this paradox requires moving beyond passive informational campaigns. Sustainable protection demands a synthesized socio-technical strategy combining experiential university training, context-sensitive in-app friction, and rapid forensic reporting via the 1930 CFCFRMS framework.

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